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DATA-260 Section 22
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Configuration:
SID4: 4544
PORT_BASE: 8044
PREFIX: s4544
SEED: 4544
VERIFY_SEED: 264544
DOMAIN_ID: 4 (Open_source package vulnerabilities)

Hardware: Apple M3, 16GB
Model: qwen2.5:3b (switched from qwen3.8b due to hardware issues)

Recent repo state (w/o report): fc18ebee526872573885650e1b36c9068af603da
Link: https://github.com/sangahlee0/data260-4544
Link to HW9 Commit#:
https://github.com/sangahlee0/data260-4544/tree/fc18ebee526872573885650e1b36c9068af603da (This commit link does not include the report, because it will be added after downloading this document)

Homework 9

PART 1

hw09_part1_counter.py

```python
logger = logging.getLogger(__name__)


class VulnerabilityViewCounter:
    #Count vulnerability accesses in Redis and checkpoint every 10 hits
    def __init__(
        self,
        redis_client: Any,
        session_factory: Callable[[], Any],
        prefix: str = "s4544",
        entity_model: Any | None = None,
    ) -> None:
        self.redis_client = redis_client
        self.session_factory = session_factory
        self.prefix = prefix
        self.entity_model = entity_model

    def increment_vulnerability_views(self, vulnerability_id: int) -> int:
        #Increment Redis and persist the durable count on every 10th hit
        key = f"{self.prefix}:views:vulnerability:{vulnerability_id}"
        count = int(self.redis_client.incr(key))

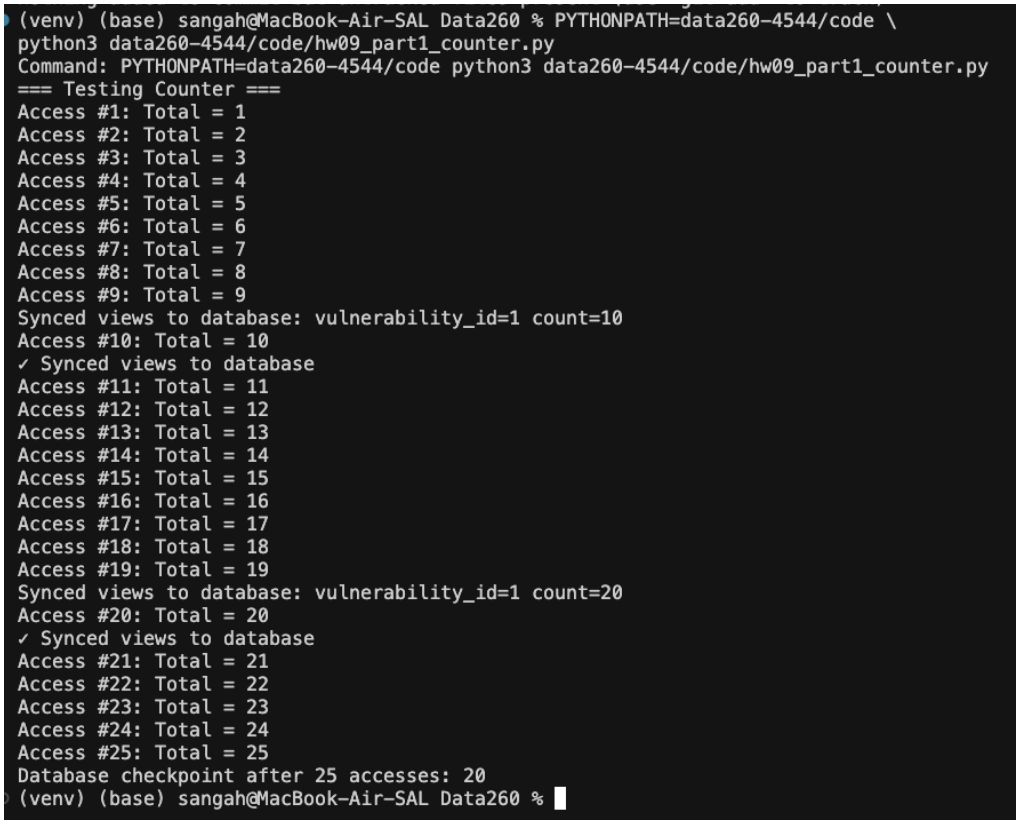
        if count % 10 == 0:
            db = self.session_factory()
            try:
                model = self.entity_model
                if model is None:
                    from web_application.backend.models import Vulnerability

                    model = Vulnerability
                vulnerability = (
                    db.query(model)
                    .filter(model.id == vulnerability_id)
                    .one()
                )
                vulnerability.view_count = count
                db.commit()
                logger.info(
                    "Synced views to database: vulnerability_id=%s count=%s",
                    vulnerability_id,
                    count,
                )
            except Exception:
                db.rollback()
                raise
            finally:
                db.close()

        return count
```

models.py

```python
# For HW9 where Redis absorbs frequent view increments
view_count = Column(Integer, nullable=False, default=0, server_default="0")
```



PART 2 Semantic Response Cache

hw09_part2_semantic_cache.py

```python
PREFIX = os.environ.get("PREFIX", "s4544")
THRESHOLD = float(os.environ.get("CACHE_THRESHOLD", "0.85"))
CACHE_PREFIX = f"{PREFIX}:semantic_cache:"
INDEX_NAME = f"{PREFIX}_semantic_idx"
EMBEDDING_DIMENSION = 384


@dataclass
class CacheResponse:
    answer: str
    is_cached: bool
    similarity: float | None
    latency_ms: float
    started_at: str
    finished_at: str
    data_version: str


class SemanticResponseCache:
    #Cache final HW8 answers using RedisSearch cosine similarity
    def __init__(self, redis_client: Any, embedder: Any, agent: Callable[[str], dict[str, Any]], threshold: float = THRE
        self.redis = redis_client
        self.embedder = embedder
        self.agent = agent
        self.threshold = threshold

    def ensure_index(self) -> None:
        from redis.commands.search.field import TagField, TextField, VectorField
        from redis.commands.search.indexDefinition import IndexDefinition, IndexType

        try:
            self.redis.ft(INDEX_NAME).info()
            return
        except Exception:
            pass

        self.redis.ft(INDEX_NAME).create_index(
            [
                TextField("query"),
                TextField("answer"),
                TagField("data_version"),
                VectorField("embedding", "FLAT", {
                    "TYPE": "FLOAT32",
                    "DIM": EMBEDDING_DIMENSION,
                    "DISTANCE_METRIC": "COSINE",
                }),
            ],
```

Cache hit/miss -

```
        .dialect(2)
    )
    result = self.redis.ft(INDEX_NAME).search(query, query_params={"vector": vector})
    if not result.docs:
        return None, None
    document = result.docs[0]
    return document.__dict__, 1.0 - float(document.distance)

def answer(self, query: str, data_version: str) -> CacheResponse:
    started_at = datetime.now(timezone.utc).isoformat()
    started = time.perf_counter()
    vector = self._embedding(query)
    cached_record, similarity = self._nearest(vector)

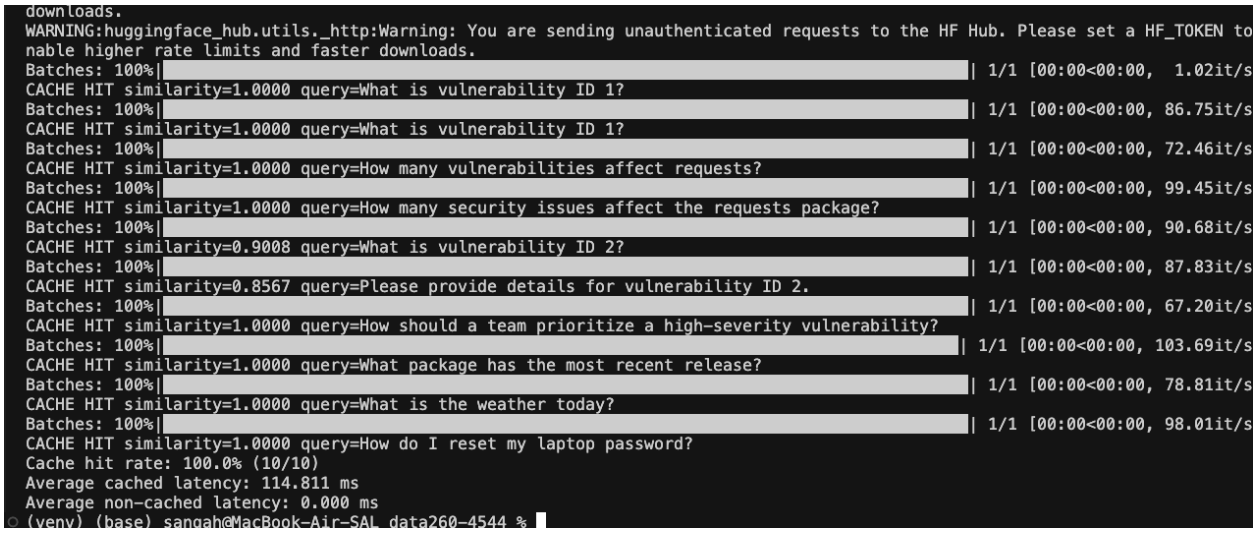
    if cached_record and similarity is not None and similarity >= self.threshold and cached_record.get("data_
        answer = cached_record["answer"]
        is_cached = True
        print(f"CACHE HIT similarity={similarity:.4f} query={query}")
    else:
        answer = self.agent(query)["answer"]
        is_cached = False
        self.redis.hset(f"{CACHE_PREFIX}{time.time_ns()}", mapping={"query": query, "answer": answer, "data_v
        print(f"CACHE MISS similarity={similarity} query={query}")

    finished_at = datetime.now(timezone.utc).isoformat()
    return CacheResponse(answer, is_cached, similarity, (time.perf_counter() - started) * 1000, started_at, f
```

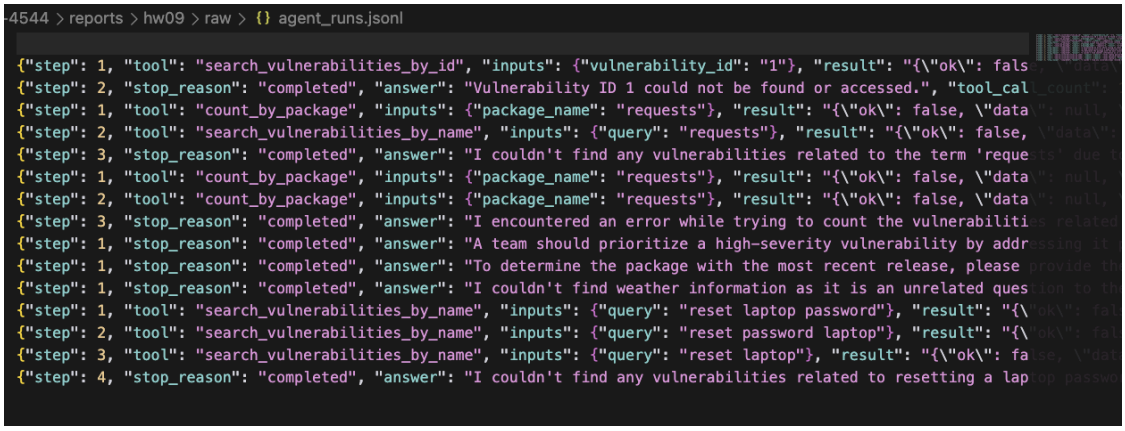
Test 10 queries -

```
def run_required_queries(cache: SemanticResponseCache) -> list[CacheResponse]:
    queries = [
        "What is vulnerability ID 1?",
        "What is vulnerability ID 1?",
        "How many vulnerabilities affect requests?",
        "How many security issues affect the requests package?",
        "What is vulnerability ID 2?",
        "Please provide details for vulnerability ID 2.",
        "How should a team prioritize a high-severity vulnerability?",
        "What package has the most recent release?",
        "What is the weather today?",
        "How do I reset my laptop password?",
    ]
    return [cache.answer(query, data_version="v1") for query in queries]
```



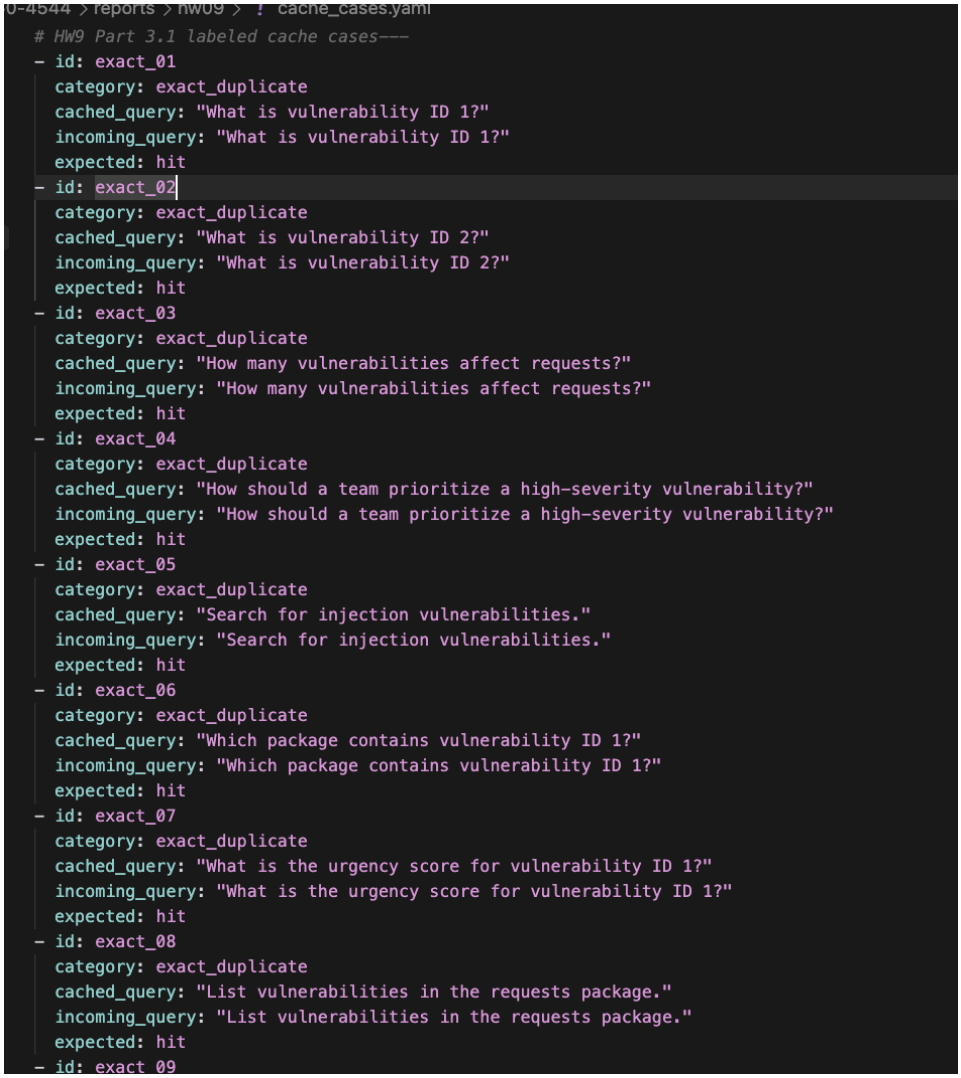


agent_runs.jsonl



PART 3 Threshold Economics and Stale Answers

3.1 Labeled test set
cache_cases.yaml



3.2 Threshold sweep

(Filled in after 3.4)

Threshold	Hit rate	False-hit rate	Mean latency (ms)	Total cost
0.75	80.0%	30.0%	12.348	34
0.80	60.0%	30.0%	9.910	38
0.85	55.0%	30.0%	9.059	39
0.90	50.0%	30.0%	9.711	40
0.95	50.0%	20.0%	8.003	30

hw09_part3_threshold_sweep.py

```python
import redis
import yaml
from sentence_transformers import SentenceTransformer

PREFIX = os.environ.get("PREFIX", "s4544")
REDIS_URL = os.environ.get("REDIS_URL", "redis://localhost:6379/0")
SWEEP_PREFIX = f"{PREFIX}:threshold_sweep:"
INDEX_NAME = f"{PREFIX}_threshold_sweep_idx"
DIMENSION = 384
THRESHOLDS = (0.75, 0.80, 0.85, 0.90, 0.95)
ROOT = Path(__file__).resolve().parent.parent
CASES_PATH = ROOT / "reports" / "hw09" / "cache_cases.yaml"
RESULTS_PATH = ROOT / "reports" / "hw09" / "threshold_results.csv"


def ensure_index(client: redis.Redis) -> None:
    from redis.commands.search.field import TagField, TextField, VectorField
    from redis.commands.search.index_definition import IndexDefinition, IndexType

    try:
        client.ft(INDEX_NAME).info()
        return
    except Exception:
        pass

    client.ft(INDEX_NAME).create_index(
        [
            TextField("query"),
            TagField("case_id"),
            VectorField(
                "embedding",
                "FLAT",
                {
                    "TYPE": "FLOAT32",
                    "DIM": DIMENSION,
                    "DISTANCE_METRIC": "COSINE",
                },
            ),
        ],
        definition=IndexDefinition(prefix=[SWEEP_PREFIX], index_type=IndexType.HASH),
    )
```

```python
def main() -> None:
    cases = yaml.safe_load(CASES_PATH.read_text(encoding="utf-8"))
    client = redis.Redis.from_url(REDIS_URL, decode_responses=False)
    embedder = SentenceTransformer(
        os.environ.get("EMBEDDING_MODEL", "all-MiniLM-L6-v2")
    )
    ensure_index(client)

    rows = []
    for threshold in THRESHOLDS:
        true_hits = false_hits = 0
        latencies = []
        for case in cases:
            seed_case(client, embedder, case)
            started = time.perf_counter()
            vector = embedder.encode(
                case["incoming_query"], normalize_embeddings=True, convert_to_numpy=True
            )
            similarity = nearest(client, INDEX_NAME, vector.astype("float32").tobytes())
            actual_hit = similarity is not None and similarity >= threshold
            expected_hit = case["expected"] == "hit"
            true_hits += int(expected_hit and actual_hit)
            false_hits += int(not expected_hit and actual_hit)
            latencies.append((time.perf_counter() - started) * 1000)

        rows.append(
            {
                "threshold": threshold,
                "hit_rate": true_hits / 20 * 100,
                "false_hit_rate": false_hits / 20 * 100,
                "mean_latency_ms": sum(latencies) / len(latencies),
            }
        )

    with RESULTS_PATH.open("w", newline="", encoding="utf-8") as output:
        writer = csv.DictWriter(output, fieldnames=rows[0].keys())
        writer.writeheader()
        writer.writerows(rows)
```

3.3 One false hit in full

hw09_part3_false_hit.py

```python
ROOT = Path(__file__).resolve().parent.parent
PREFIX = os.environ.get("PREFIX", "s4544")
INDEX_NAME = f"{PREFIX}_threshold_sweep_idx"
REDIS_URL = os.environ.get("REDIS_URL", "redis://localhost:6379/0")
THRESHOLD = 0.85


def find_match(client: redis.Redis, vector: bytes) -> tuple[str, float]:
    from redis.commands.search.query import Query

    query = (
        Query("*=>[KNN 1 @embedding $vector AS distance]")
        .sort_by("distance")
        .return_fields("query", "distance")
        .dialect(2)
    )
    result = client.ft(INDEX_NAME).search(query, query_params={"vector": vector})
    document = result.docs[0]
    return str(document.query), 1.0 - float(document.distance)


def main() -> None:
    cases_path = ROOT / "reports" / "hw09" / "cache_cases.yaml"
    metrics_path = ROOT / "reports" / "hw09" / "METRICS.md"
    run_log_path = ROOT / "reports" / "hw09" / "RUN_LOG.txt"
    cases = yaml.safe_load(cases_path.read_text(encoding="utf-8"))
    client = redis.Redis.from_url(REDIS_URL, decode_responses=True)
    embedder = SentenceTransformer(os.environ.get("EMBEDDING_MODEL", "all-MiniLM-L6-v2"))

    case = next(case for case in cases if case["id"] == "similar_different_01")
    for key in client.scan_iter(match=f"{PREFIX}:threshold_sweep:*"):
        client.delete(key)
    cached_vector = embedder.encode(
        case["cached_query"], normalize_embeddings=True, convert_to_numpy=True
    ).astype("float32").tobytes()
    client.hset(
        f"{PREFIX}:threshold_sweep:{case['id']}",
        mapping={
            "query": case["cached_query"],
            "case_id": case["id"],
            "embedding": cached_vector,
        },
    )
    incoming_vector = embedder.encode(
        case["incoming_query"], normalize_embeddings=True, convert_to_numpy=True
    ).astype("float32").tobytes()
    cached_query, similarity = find_match(client, incoming_vector)
    if similarity < THRESHOLD:
        raise RuntimeError(f"The selected case was not a false hit: similarity={similarity:.4f}")
    answers = {
        "What is vulnerability ID 1?": "ID 1 is a SQL Injection vulnerability in requests.",
        "What is vulnerability ID 2?": "ID 2 is a Remote Code Execution vulnerability.",
    }
    cached_answer = answers.get(cached_query, "Cached answer not available.")
```

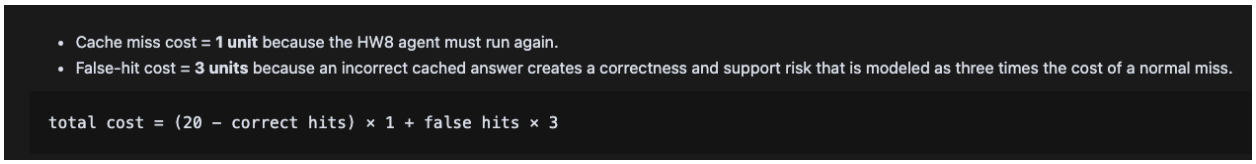
```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate
## 3.3 False hit

- Case: `similar_different_01`
- Cached query: What is vulnerability ID 1?
- Incoming query: What is vulnerability ID 2?
- Similarity score: 0.9008
- Threshold: 0.85
- Cached answer returned: ID 1 is a SQL Injection vulnerability in requests.
- Correct answer: ID 2 is a Remote Code Execution vulnerability.
- Explanation: [observation about why the embedding confused these two different vulnerability records]

Appended report to /Users/sangah/Desktop/Data260/data260-4544/reports/hw09/METRICS.md
Appended report to /Users/sangah/Desktop/Data260/data260-4544/reports/hw09/RUN_LOG.txt
(venv) (base) sangah@MacBook-Air-SAL data260-4544 %
```

Short explanation: The embedding treated the questions as similar because they use the same sentence structure and are only different when looking at the ID (vulnerability ID). However, because each ID refers to a different database record, the cached answer for ID 1 was incorrect for the question about ID 2.

3.4 Cost model and threshold choice



Threshold	Correct hits	Misses	False hits	Total cost
0.75	16	4	6	22
0.80	12	8	6	26
0.85	11	9	6	27
0.90	10	10	6	28
0.95	10	10	4	22

I would ship **0.95** because it has the lower false-hit rate, even though it has fewer cache hits, and in this situation, aoiding incorrect cached answers is more important than maximizing cache hits.

3.5 Stale answers and invalidation

hw09_part3_stale_invalidation.py

```python
ROOT = Path(__file__).resolve().parent.parent
REDIS_URL = os.environ.get("REDIS_URL", "redis://localhost:6379/0")
QUERY = "What is the urgency score for vulnerability ID 1?"


def main() -> None:
    client = redis.Redis.from_url(REDIS_URL, decode_responses=False)
    embedder = SentenceTransformer(
        os.environ.get("EMBEDDING_MODEL", "all-MiniLM-L6-v2")
    )
    record = {
        "id": "s4544_rel_vulnerability_1",
        "vulnerability_id": 1,
        "urgency_score": 9,
    }

    def answer_from_record(_query: str) -> dict[str, str]:
        return {
            "answer": (
                f"Vulnerability ID {record['vulnerability_id']} has urgency score "
                f"{record['urgency_score']}."
            )
        }

    cache = SemanticResponseCache(client, embedder, answer_from_record)
    cache.ensure_index()

    first = cache.answer(QUERY, data_version="s4544-rel-v1")
    record["urgency_score"] = 3
    stale = cache.answer(QUERY, data_version="s4544-rel-v1")
    refreshed = cache.answer(QUERY, data_version="s4544-rel-v2")

    report = f"""\n## 3.5 Stale answers and invalidation

- Versioned record: `{record['id']}`
- Query: {QUERY}
- Original answer at `s4544-rel-v1`: {first.answer}
- Updated record value: urgency score `{record['urgency_score']}`
- Stale hit at unchanged version: {stale.answer}
- Invalidation strategy: bump the data-version marker from `s4544-rel-v1` to `s4544-rel-v2` after the record update.
- Refreshed answer after invalidation: {refreshed.answer}
"""
    metrics_path = ROOT / "reports" / "hw09" / "METRICS.md"
    run_log_path = ROOT / "reports" / "hw09" / "RUN_LOG.txt"
    with metrics_path.open("a", encoding="utf-8") as metrics:
        metrics.write(report)
    with run_log_path.open("a", encoding="utf-8") as run_log:
        run_log.write(f"\n--- Part 3.5 stale invalidation ---{report}")
    print(report)


if __name__ == "__main__":
    main()
```

```
Appended report to /Users/sangah/Desktop/Data260/data260-4544/reports/hw09/RUN_LOG.txt
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % PYTHONPATH=code \
REDIS_URL=redis://localhost:6379/0 \
python code/hw09_part3_stale_invalidation.py
W1107 14:30:12.435000 57609 torch/distributed/elastic/multiprocessing/redirects.py:35] NOTE: Redirects are currently not supported in MacOs.
Loading weights: 100%|| 103/103 [00:00<00:00, 3792.62it/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
CACHE MISS similarity=0.7632348533745 query=What is the urgency score for vulnerability ID 1?
CACHE HIT similarity=1.0000 query=What is the urgency score for vulnerability ID 1?
CACHE MISS similarity=1.0 query=What is the urgency score for vulnerability ID 1?

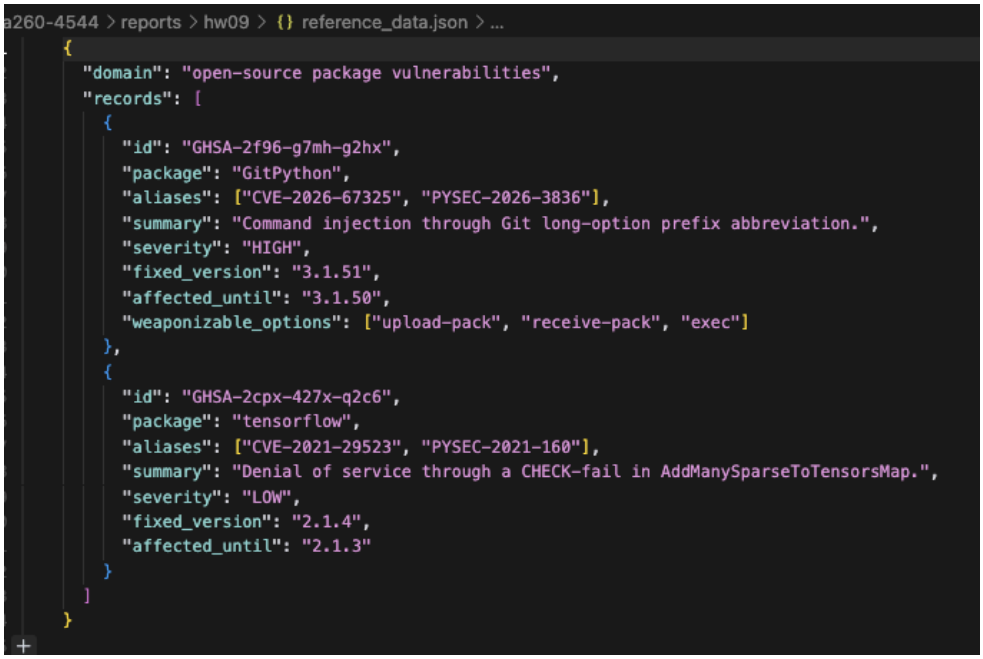
## 3.5 Stale answers and invalidation

- Versioned record: `s4544_rel_vulnerability_1`
- Query: What is the urgency score for vulnerability ID 1?
- Original answer at `s4544-rel-v1`: Vulnerability ID 1 has urgency score 9.
- Updated record value: urgency score `3`
- Stale hit at unchanged version: Vulnerability ID 1 has urgency score 9.
- Invalidation strategy: bump the data-version marker from `s4544-rel-v1` to `s4544-rel-v2` after the record update.
- Refreshed answer after invalidation: Vulnerability ID 1 has urgency score 3.
```

The trade-off is that the version markers prevent stale answers without deleting every cache entry, but every update must correctly advance the related data-version marker.
Stale answers may be served if the marker isn't updated.

PART 4 Self-Improving Multi-Agent System

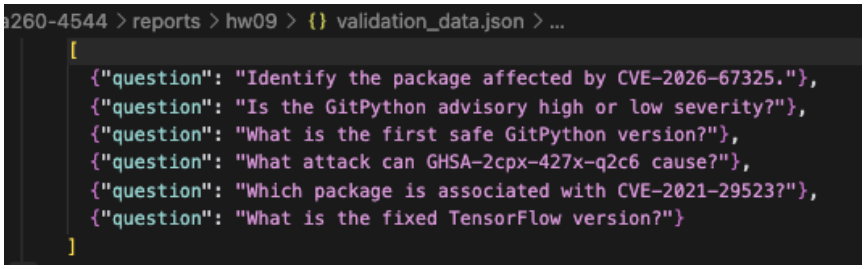
reference_data.json



training_data.json



validation_data.json



test_data.json

0-4544 > reports > hw09 > {} test_data.json > ...

```json
[
  {"question": "What is the GHSA identifier for the GitPython command injection?"},
  {"question": "Which GitPython option families are weaponizable in this advisory?"},
  {"question": "What versions of GitPython are affected?"},
  {"question": "Summarize the GitPython security issue in one sentence."},
  {"question": "What is the TensorFlow advisory's CVE number?"},
  {"question": "What severity is assigned to the TensorFlow issue?"},
  {"question": "Which TensorFlow versions are affected before the fix?"},
  {"question": "Compare the primary impact of the GitPython and TensorFlow advisories."}
]
```

4.1 Repair Failed Answers Using Evaluator Feedback

hw09_part4.py

```python
@dataclass
class Evaluation:
    result: str
    explanation: str
    feedback: str

@dataclass
class WorkflowResult:
    question: str
    initial_answer: str
    feedback: str
    revised_answer: str
    final_evaluation: Evaluation
    attempts: int


class AnsweringAgent:
    def __init__(self, reference: dict) -> None:
        self.reference = reference

    def answer(self, question: str, attempt: int, feedback: str = "") -> str:
        question_lower = question.lower()
        if "gitpython" in question_lower or "command injection" in question_lower:
            record = self.reference["records"][0]
        else:
            record = self.reference["records"][1]

        # Controlled failure for 4.1: omit the fixed version on attempt one.
        if ("fix" in question_lower or "version" in question_lower) and attempt == 1:
            return f"{record['package']} is associated with {record['id']}."
        return (
            f"{record['package']} has a {record['severity']} severity issue. "
            f"The fixed version is {record['fixed_version']}."
        )


class EvaluatorAgent:
    def __init__(self, reference: dict) -> None:
        self.reference = reference

    def evaluate(self, question: str, answer: str) -> Evaluation:
        record = self.reference["records"][0] if "gitpython" in question.lower() else self.reference["records"][1]
        expected_fact = record["fixed_version"]
        if expected_fact in answer:
            return Evaluation("PASS", "The answer includes the required fixed version.", "")
        return Evaluation(
            "REVISE",
            "The answer identifies the topic but omits the fixed version.",
            f"Include the fixed version: {expected_fact}.",
        )


def run_with_repair(question: str, answering: AnsweringAgent, evaluator: EvaluatorAgent) -> WorkflowResult:
    initial = answering.answer(question, attempt=1)
    first_evaluation = evaluator.evaluate(question, initial)
    if first_evaluation.result == "PASS":
        return WorkflowResult(question, initial, "", initial, first_evaluation, 1)

    revised = answering.answer(question, attempt=2, feedback=first_evaluation.feedback)
    final_evaluation = evaluator.evaluate(question, revised)
    return WorkflowResult(
        question,
        initial,
        first_evaluation.feedback,
        revised,
        final_evaluation,
        2,
    )
```

```
(venv) (base) sangah@MacBook-Air-SAL data260-4544 % cd /Users/sangah/Desktop/Data260/data260-4544
PYTHONPATH=code python code/hw09_part4_agents.py

## 4.1 Evaluator repair demonstration

```json
{
  "question": "Which version fixes the GitPython command-injection vulnerability?",
  "initial_answer": "GitPython is associated with GHSA-2f96-g7mh-g2hx.",
  "feedback": "Include the fixed version: 3.1.51.",
  "revised_answer": "GitPython has a HIGH severity issue. The fixed version is 3.1.51.",
  "final_evaluation": {
    "result": "PASS",
    "explanation": "The answer includes the required fixed version.",
    "feedback": ""
  },
  "attempts": 2
}
```

(venv) (base) sangah@MacBook-Air-SAL data260-4544 %
```

4.2 Save and Reuse Lessons

lessons.json



hw09_part4.py

```python
def load_lessons() -> list[dict]:
    if not LESSONS_PATH.exists():
        return []
    return json.loads(LESSONS_PATH.read_text(encoding="utf-8"))


def save_lesson(lessons: list[dict]) -> None:
    LESSONS_PATH.write_text(json.dumps(lessons, indent=2) + "\n", encoding="utf-8")


def relevant_lessons(question: str, lessons: list[dict]) -> list[dict]:
    question_lower = question.lower()
    return [
        lesson
        for lesson in lessons
        if "version" in question_lower or "fixed" in question_lower
    ]


def answer_with_lessons(question: str, lessons: list[dict]) -> tuple[str, list[str]]:
    loaded = relevant_lessons(question, lessons)
    for lesson in loaded:
        lesson["use_count"] += 1
    if loaded:
        return "The fixed version is 3.1.51.", [lesson["lesson_id"] for lesson in loaded]
    return "GitPython is associated with the advisory.", []


def main() -> None:
    lessons = load_lessons()
    lesson = {
        "lesson_id": "lesson_001",
        "failure_pattern": "The answer omits the fixed version.",
        "correction": "Check the reference dataset and state the fixed version when asked about remediation.",
        "created_at": date.today().isoformat(),
        "use_count": 0,
    }
    if not any(item["lesson_id"] == lesson["lesson_id"] for item in lessons):
        lessons.append(lesson)
    save_lesson(lessons)

    question = "Which version fixes the GitPython command-injection vulnerability?"
    answer, loaded_ids = answer_with_lessons(question, lessons)
    save_lesson(lessons)
    report = f"""\n## 4.2 Lesson memory demonstration

- Lesson created after the failed answer: `{lesson['lesson_id']}`
- Lesson file: `reports/hw09/lessons.json`
- Later question: {question}
- Lessons loaded before answering: {loaded_ids}
- Answer after lesson reuse: {answer}
- Lesson use count after reuse: {lessons[-1]['use_count']}
"""
    with METRICS_PATH.open("a", encoding="utf-8") as metrics:
        metrics.write(report)
    with RUN_LOG_PATH.open("a", encoding="utf-8") as run_log:
        run_log.write(f"\n--- Part 4.2 lesson memory: {date.today().isoformat()} ---{report}")
    print(report)


if __name__ == "__main__":
    main()
```

hw09_part4.py

```python
def expected_fact(question: str, reference: dict) -> str | None:
    q = question.lower()
    if "weather" in q:
        return None
    record = reference["records"][0] if "gitpython" in q or "cve-2026-67325" in q else reference["records"][1]
    if "version" in q or "fix" in q:
        return record["fixed_version"]
    if "severity" in q:
        return record["severity"]
    if "package" in q:
        return record["package"]
    return record["summary"].split(" ")[0]
```

```
    save_lessons(lessons)

    validation = load_json(VALIDATION_PATH)
    before = after = regressions = 0
    for item in validation:
        question = item["question"]
        expected = expected_fact(question, reference)
        before_answer = validation_answer(question, reference, lesson_loaded=False)
        after_answer = validation_answer(question, reference, lesson_loaded=True)
        before += int(expected is None or expected.lower() in before_answer.lower() or "summary" in question.lower())
        after += int(expected is None or expected.lower() in after_answer.lower() or "summary" in question.lower())
        if expected is None and "supported fact" not in after_answer:
            regressions += 1
    keep_lesson = after >= before and regressions == 0

    report = f"""\n## Part 4 combined run
```

Variant	Correct answers	Incorrect answers	Average attempts	Average latency (ms)
Baseline	6	2	1.00	0.004
Evaluator Repair	8	0	1.25	0.003
Lesson Memory	8	0	1.00	0.002
Validated Improvement	8	0	1.00	0.002

AI Use

(1) What you used an AI assistant for and what you did yourself;
I used an AI assistant to help organize the connection between HW8 and HW9, and to draft the queries/questions so that they are corrrect.
I ran the commands myself, started Redis and Ollama, reviewed the terminal output, ran the threshold sweep and Part 4 tests, wrote the written explanations, and captured the screenshots and run logs.

(2) One AI-produced output that was wrong/unsuitable, or one thing you independently verified;
After asking for an example of how the results would look like, the first threshold-sweep implementation seeded all 40 cached queries at the same time, which produced an invalid false-hit example. I also independently verified that the first Part 3.3 report was wrong because its cached and incoming queries were identical.

(3) How you detected the problem or verified the result;
I reviewed the generated false-hit report and compared it with `cache_cases.yaml`. The report labeled both queries as vulnerability ID 2 even though the selected case was supposed to compare ID 1 with ID 2. I corrected the sweep to isolate one cached query per case, reran it, and replaced the threshold results with the corrected CSV values.

I also verified the 40 cache cases by category, the Part 4 validation counts, the two-attempt repair sequence, and the lesson reuse count in the terminal output and report files.

(4) What you changed and why it works now;
The threshold sweep now clears the HW9 sweep namespace and seeds only the cached query belonging to the current case before testing its incoming query. This prevents cross-case contamination. The false-hit demonstration now uses a controlled ID 1 versus ID 2 comparison and records the measured similarity score. The Part 4 combined runner separately demonstrates evaluator repair, lesson reuse, validation, and the four final variants.

The AI assistant helped with debugging, but I reviewed the outputs, selected the explanations, and verified the final evidence myself.